

Arjun Pillai

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RESEARCH EXPERIENCE

Research Internship

Stanford Artificial Intelligence Lab

May 2026 - Present

- Worked under Dr. Chen Shani under the Stanford Natural Language Processing Group
- Designed and implemented a fine-tuning pipeline comparing aggregated (majority-vote) vs. non-aggregated (per-annotator) label training strategies across two datasets: a vaccine hesitancy stance classification dataset and ChaosNLI (an NLI dataset with 100 annotations per example)
- Fine-tuned and evaluated 10 transformer models spanning encoder (BERT-base, RoBERTa-base, ModernBERT, NeoBERT, DeBERTa-v3-large) and decoder (GPT-2, SmolLM2-135M, SmolLM2-360M, Pythia-160M, GPT-X2-125M) architectures
- Implemented 5-fold stratified cross-validation for robust evaluation, with per-fold random annotator sampling for ChaosNLI to maximize use of its annotation richness
- Identified training instability in RoBERTa under non-aggregated labels (macro F1 std dev ± 0.21 vs. ± 0.03 for all other configs), and found that non-aggregated training did not consistently improve performance for either encoder or decoder models
- Built a modular codebase with a shared utility library covering data loading, CV loop, dataset classes, and metrics, supporting both datasets and all models from a single interface

Student Researcher

MIT Critical Data

July 2026 - Present

- Co-designed a pre-registered, factorial prompt experiment auditing how LLMs disclose the provenance of medical knowledge they generate, with independent verification of every returned citation.
- Built a codebook from scratch to classify AI-generated citations as fabricated, real-with-bibliographic-errors, or fully accurate, iterating the category rules against 10-15 pilot citations before freezing it.
- Ran a 4-tier escalating prompt protocol (plain question $\rightarrow$ sourcing request $\rightarrow$ provenance interrogation $\rightarrow$ adversarial pushback) across multiple LLMs on adolescent mental health questions, logging model version, run date, and retrieval setting for every session.
- Manually verified all returned citations against PubMed, Crossref, and Google Scholar to compute fabrication and bibliographic-error rates, without AI assistance in the coding process.
- Conducted independent double-coding with a teammate and computed inter-rater reliability (Cohen's kappa), with a third teammate adjudicating disagreements below the reliability threshold.

Regression Research Lead

Independent Research

December 2025 - June

2026

- Developed and implemented OLS, Ridge, and Lasso regression models from first principles using NumPy to predict building thermal efficiency.
- Derived the Normal Equation for closed-form solutions and implemented Coordinate Descent for Lasso regularization to handle feature collinearity.

- Achieved an accuracy of 90.8%, matching industry-standard scikit-learn benchmarks while ensuring 100% mathematical transparency in the codebase.
 - Analyzed a factorial design dataset of 768 building configurations, evaluating the impact of 8 geometric features on primary energy consumption.
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PROJECTS

Lyze: Multimodal RAG Intelligence System

November 2025 - Present

- Built a multimodal RAG pipeline using FastAPI and Qdrant, enabling semantic search across PDFs, images, audio, and video in a unified 768-dimensional vector store.
 - Engineered a multimodal ingestion system using LLaVA for image captioning and Whisper for audio/video transcription, supporting cross-modal querying across 4 file types in a single search.
 - Designed a provider-agnostic architecture supporting 4 AI backends (Ollama, OpenAI, Anthropic, Gemini) with a React + TypeScript frontend and Python REST API.
 - Garnered over 2.2k views across platforms around the world in the United States, India, United Kingdom, and more.
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SKILLS

- **Programming:** C++, Java, Python, Typescript, HTML, MATLAB
 - **Technical:** Machine Learning, Linear Regression, Data Analyst, Language Interpretability,
 - **Frameworks/Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
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AWARDS

- **IAI²O Finalist** - Represented the US on an international scale, placed 2/54 US Finalists in Regionals
- **SMT 2025 Distinguished Honorable Mention** - Top 20% in Team + Power Round
- **BMT 2025 Honorable Mention** - Top 50% in Algebra Round
- **SMT 2026 Distinguished Honorable Mention** - Top 20% in Team + Power Round